Multiband Attention Network for Motor Imagery EEG Classification

Uddeshay Anand[1], Kusum Kumari Bharti[2] and Mohd Sameer[3]

Department of Information Technology
Dr B R Ambedkar National Institute of Technology, Jalandhar, India
uddeshaya.da.24@nitj.ac.in, bhartik@nitj.ac.in, mohds.da.24@nitj.ac.in

Abstract. Motor imagery (MI)-based brain-computer interfaces (BCIs) offer non-invasive brain-to-device communication, demonstrating potential for neurorehabilitation and assistive technology. Accurately classifying MI electroencephalography (EEG) signals remains challenging due to their inherently low signal-to-noise ratio, non-stationarity and substantial inter-subject variability. Another challenge is the limited availability of labeled EEG data, making it difficult to effectively train deep learning models. An end-to-end deep learning architecture, termed the Multiband Attention Network (MB-ATNet), is proposed to address the identified issues. The model utilizes frequency-specific spectral streams (theta, alpha and beta) at the input level. It further integrates a Convolutional Block Attention Module (CBAM) to enable joint channel and spatial feature recalibration. Data augmentation is used to improve the generalization of the proposed model. BCI Competition IV-2a (BCIC-IV-2a) dataset was used to evaluate performance across nine subjects. MB-ATNet shows an overall mean classification accuracy of 77.17% and mean Cohen's kappa of 0.696, outperforming established baselines on individual subjects while demonstrating competitive overall performance. These results suggest that combining multiband spectral fusion with dual-axis attention leads to more robust MI-EEG decoding.

Keywords: Brain-computer interface · Electroencephalogram · Motor imagery · Multiband fusion · Attention mechanism · Data augmentation · Deep learning

1 Introduction

Brain-computer interfaces (BCIs) provide a direct communication channel between the brain and external devices by translating neural activity into control commands, eliminating the need for muscular activity [1]. People with severe motor impairments, such as prosthetic limb control, wheelchair navigation and post stroke or spinal cord injury neurological rehabilitation, have critical applications of BCI systems [2].

EEG is the most common non-invasive brain recording method for BCIs as it offers high temporal resolution, portability and low cost [3]. Other modalities such as magnetoencephalography (MEG), functional near-infrared spectroscopy

(fNIRS) and functional magnetic resonance imaging (fMRI) provide comparable spatial resolution but are often limited by complex hardware requirements and strict motion constraints, making practical deployment difficult [3]. EEG-based BCIs take advantage of event related desynchronization (ERD) and event related synchronization (ERS) phenomena that occur in the alpha (8–13 Hz) and beta (13–30 Hz) bands of the sensorimotor cortex while imagining limb movements [4].

Common Spatial Pattern (CSP) algorithm and Filter Bank CSP (FBCSP) are used to extract features in traditional MI-BCI pipelines [5]. Despite their effectiveness, these methods require domain expertise, are sensitive to noise and artifacts and neglect temporal dynamics in variance-based feature computation. Linear Discriminant Analysis (LDA), Support Vector Machines (SVM) and Relevance Vector Machines (RVM) classifiers have limited representational capacity when trained on shallow and handcrafted feature vectors [6]. Phase-space reconstruction methods (PSCSP) enhance classical CSP by projecting EEG into a high-dimensional phase space before feature extraction, but still require manual preprocessing overhead and plateau in performance [6]. Signal processing plays a foundational role in MI-BCI pipelines: effective time-frequency decomposition of non-stationary EEG signals enables extraction of band-specific neural dynamics that are otherwise obscured in the raw signal.

Deep learning has enhanced MI-EEG decoding by allowing automatic extraction of hierarchical features from minimally preprocessed signals. Convolutional neural networks (CNNs) capture spatial and spectral patterns through local receptive fields [7] whereas temporal convolutional networks (TCNs) extend temporal context using dilated causal convolutions without suffering from vanishing gradient issues [2]. CBAM attention mechanisms are used to selectively recalibrate informative feature dimensions [8] and multi-branch architectures use parallel streams of different kernel scales to enrich spectral-temporal representations [2, 3]. The lack of labeled EEG data is one of the main issues as each subject only provides 288 training trials across four classes. To solve this, CutCat augmentation [9] splices quarter-length segments of same-class trials to generate new instances, adding inter-trial variation and promoting class-discriminative rather than subject-specific learning.

The rest of the paper is structured as follows. Section 2 presents the proposed MB-ATNet model including the data augmentation strategy, multi-band spectral decomposition and the detailed architecture including the attention block. Section 3 outlines the experimental setup, dataset description, preprocessing pipeline, training strategy, hyperparameter configuration and evaluation metrics. Section 4 presents the classification results, performance analysis and a comparative evaluation with state-of-the-art methods. Section 5 concludes the paper with a summary of findings, practical implications, limitations and directions for future research.

2 Proposed Methodology

The MB-ATNet pipeline consists of three sequential stages: (1) CutCat data augmentation, (2) multi-band spectral filtering and (3) the MB-ATNet classification network.

2.1 CutCat Augmentation

To address the inherent data scarcity in MI-EEG datasets, the CutCat augmentation strategy proposed by Al-Saegh *et al.* [9] is employed. Given two training trials $\mathbf{X}_a, \mathbf{X}_b \in \mathbb{R}^{C\times T}$ belonging to the same class label y, a synthetic trial $\mathbf{X}_{new}$ is formed by interleaving quarter-length temporal segments:

$$\mathbf{X}_{new} = \big[\mathbf{X}_a(:,\, 0:q),\ \mathbf{X}_b(:,\, q:2q),\ \mathbf{X}_a(:,\, 2q:3q),\ \mathbf{X}_b(:,\, 3q:T)\big], \qquad (1)$$

where $q = T/4$ is the quarter-segment length and $[\cdot]$ represents temporal concatenation. With $T = 1000$ samples (4 s at 250 Hz), each segment spans 250 time points. The synthetic trial $\mathbf{X}_{new}$ inherits the class label y of both source trials. This splicing strategy is applied exclusively to the training set; validation and test splits remain untouched. Each original trial contributes to multiple synthetic trials, substantially increasing the effective training corpus and exposing the model to diverse intra-class temporal patterns.

2.2 Multi-Band Spectral Decomposition

Signal preprocessing is a critical step in EEG-based BCI pipelines because neural recordings are inherently contaminated by noise, artifacts and overlapping physiological rhythms. Classical time-frequency methods such as STFT and Wavelet Transform have fixed resolution trade-offs and need parameter tuning, which restricts their use in automated deep learning pipelines. Band-pass Butterworth filtering provides a computationally efficient and physiologically grounded alternative, directly isolating neural oscillatory bands relevant to motor imagery [10].

EEG signals are split into four parallel streams: raw (full-spectrum, DC offset removed), theta (4–8 Hz), alpha (8–13 Hz) and beta (13–30 Hz). Each stream is obtained using a 4th-order zero-phase Butterworth filter applied via `filtfilt` to avoid group-delay distortion. The DC offset of each trial is removed prior to filtering by subtracting the temporal mean. All filtering is performed with channels in the last dimension (N, T, C), where N is the batch size, $T = 1000$ time points and $C = 22$ EEG channels, with streams computed independently from the augmented training data.

The theta band encodes cognitive processes, while the alpha band carries sensorimotor idle rhythms that desynchronize contralaterally during MI and the beta band reflects post-movement synchronization relevant to motor intention [3]. Delta activity (0.5–4 Hz) predominantly reflects slow cortical potentials unrelated to voluntary movement, while gamma-band responses, although present during

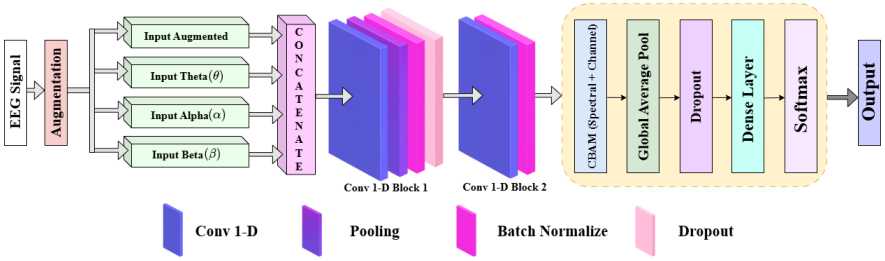


Fig. 1. Architecture of the proposed Multiband Attention Network model.

MI, are highly susceptible to muscular artifacts conflicting with the end-to-end design of MB-ATNet [10]. Together, theta, alpha and beta rhythms encode the most discriminative dynamics for MI decoding [11] and feeding all four streams simultaneously allows MB-ATNet to learn cross-band interaction patterns without designer-imposed frequency restrictions [7]. CBAM is preferred over standard squeeze-and-excitation (SE) blocks as its dual-axis design recalibrates both channel and spatial-temporal dimensions, allowing selective focus on informative frequency channels and discriminative time segments within each MI trial [12].

2.3 MB-ATNet Architecture

The MB-ATNet architecture is illustrated in Fig. 1. It accepts four inputs of shape $(N, 1000, 22)$, consisting of raw signal, theta, alpha and beta band components which are early fused by concatenation along the channel axis to produce a tensor of shape $(N, 1000, 88)$.

Convolutional Feature Extraction. The fused tensor passes through two successive 1-D convolutional blocks. The first block applies 32 filters with a kernel size of 5, ReLU activation, same-padding, followed by MaxPooling1D(2) and Batch Normalization (BN). A Dropout layer with rate 0.4 is appended for regularization. The second block applies 64 filters with a kernel size of 3, ReLU activation, same-padding and BN. Formally, for the l-th convolutional layer:

$$\mathbf{H}^{(l)} = \phi\Big(\mathrm{BN}\Big(\mathbf{W}^{(l)} * \mathbf{H}^{(l-1)} + \mathbf{b}^{(l)}\Big)\Big), \tag{2}$$

where $*$ denotes 1-D convolution, $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are learnable weights and biases, $\phi(\cdot)$ is ReLU activation and BN normalizes the pre-activation feature map.

CBAM Attention Block. Following the second convolutional block, a 1-D adaptation of the Convolutional Block Attention Module (CBAM) [12] is applied. CBAM sequentially performs channel attention followed by spatial

attention, guiding the model on *what* features to emphasize and *where* along the time axis to focus. CBAM is selected over simpler channel-only attention mechanisms (e.g., squeeze-and-excitation networks) because its dual-axis design, which sequentially recalibrates both channel and spatial dimensions, is better suited to the heterogeneous spectral and temporal structure of multiband EEG inputs [8, 12].

Channel Attention. Given the feature map $\mathbf{F} \in \mathbb{R}^{T' \times C'}$, channel attention is computed by aggregating temporal context through both global average pooling (GAP) and global max pooling (GMP):

$$\mathbf{M}_c = \sigma(\mathrm{MLP}(\mathrm{GAP}(\mathbf{F})) + \mathrm{MLP}(\mathrm{GMP}(\mathbf{F}))), \tag{3}$$

where the shared MLP has reduction ratio $r = 8$ and $\sigma(\cdot)$ is the sigmoid function. The channel-attended feature map is $\mathbf{F}' = \mathbf{M}_c \odot \mathbf{F}$.

Spatial Attention. Spatial attention is applied to $\mathbf{F}'$ by computing average- and max-pooled descriptors along the channel dimension, concatenating them and convolving with a 1-D filter of kernel size 7:

$$\mathbf{M}_s = \sigma\big(f^{7\times1}\big(\big[\mathrm{AvgPool}_c(\mathbf{F}'),\ \mathrm{MaxPool}_c(\mathbf{F}')\big]\big)\big), \tag{4}$$

where $f^{7\times1}$ denotes a 1-D convolution with a 7-sample kernel, same-padding and sigmoid activation. The final attended feature map is $\mathbf{F}'' = \mathbf{M}_s \odot \mathbf{F}'$. A residual connection $\mathbf{F}'' = \mathbf{F}'' + \mathbf{F}$ is applied optionally to stabilize gradient flow.

The significance of CBAM recalibration is closely tied to the multiband input structure of MB-ATNet. Following early concatenation of the four spectral streams, the fused feature map encodes heterogeneous rhythmic content across 88 channels (22 electrodes $\times$ 4 bands). The channel attention module selectively suppresses non-task-relevant frequency channels, such as raw signal components during low signal-to-noise periods or theta channels in subjects with weak cognitive engagement, while amplifying alpha and beta channels that exhibit stronger contralateral ERD during active motor imagery [10]. The spatial attention module complements this by identifying the most discriminative temporal segments within the 4-s MI window; since ERD typically peaks between 1–3 s post-cue [3], spatial attention downweights pre- and post-movement intervals that contribute noise rather than discriminative information. Together, the two attention axes ensure that band-specific and temporally localized features are jointly recalibrated before classification.

Classification Head. The attended feature map is reduced by Global Average Pooling (GAP) to a 64-dimensional descriptor, followed by a Dropout layer (rate 0.3). A fully connected layer with 64 units and L2 regularization ($\lambda = 0.002$) is followed by a four-way softmax output layer corresponding to the four MI classes (left hand, right hand, foot, tongue). The complete architecture of MB-ATNet is summarised in Table 1, with a total of 26,234 trainable parameters, making it computationally lightweight compared to architectures such as TCNet-Fusion and EEG-FMCNN.

Table 1. MB-ATNet architecture summary.

Component	Configuration	Output shape
Input fusion	Concat(raw, θ, α, β)	$(N, 1000, 88)$
Conv1D Block 1	32 filters, k=5, MaxPool(2), BN, Drop(0.4)	$(N, 500, 32)$
Conv1D Block 2	64 filters, k=3, BN	$(N, 500, 64)$
CBAM (Ch.)	Shared MLP, r=8, GAP+GMP	$(N, 500, 64)$
CBAM (Sp.)	Conv1D k=7, sigmoid	$(N, 500, 64)$
GAP	Global Average Pooling	$(N, 64)$
Dense	64 units, L2(λ=0.002), Drop(0.3)	$(N, 64)$
Output	Softmax, 4 classes	$(N, 4)$

3 Experimental Setup

3.1 Dataset

MB-ATNet is evaluated on the BCI Competition IV-2a Dataset (BCIC-IV-2a) [13], a widely used benchmark for four-class MI-EEG classification. The dataset comprises recordings from nine subjects across two sessions: Session T for training and Session E for evaluation. Each session contains 288 trials equally distributed across four classes: left-hand MI, right-hand MI, foot MI and tongue MI (72 trials per class). EEG was recorded from 22 electrodes following the standard 10–20 system, alongside 3 EOG channels, at a sampling rate of 250 Hz with hardware band-pass filtering at 0.5–100 Hz and a 50 Hz notch filter. Each trial follows a paradigm of a 2-s fixation period, a 1.25-s directional cue and 4 s of MI task execution.

3.2 Preprocessing

Data loading follows the MATLAB file structure of BCIC-IV-2a. For each trial, the segment from $t = 2$ s to $t = 6$ s (motor imagery window) is extracted, yielding a trial tensor of shape 22×1000 (channels $\times$ samples). All 288 trials per session are retained, including those flagged with artifacts in the original dataset, to avoid selection bias. The data are transposed to channels-last format $(1000, 22)$ for compatibility with TensorFlow's 1-D convolution API. No band-pass pre-filtering is applied prior to augmentation; spectral decomposition (Section 2.2) is performed after the augmentation step using the expanded training set.

3.3 Training Strategy

Following augmentation, the expanded training set is partitioned into a 90% training and 10% stratified validation split, with Session E held out for final evaluation. Training hyperparameters including the optimizer, learning rate, early-stopping threshold, regularization settings and filter configuration are summarized in Table 2. Training runs for a maximum of 1000 epochs with a custom early-stopping callback: training terminates when validation loss falls below a threshold of $\delta = 0.025$.

Table 2. MB-ATNet training hyperparameters.

Hyperparameter	Value
Optimizer	Adam
Learning rate	10^{-4}
Batch size	64
Max. epochs	1000
Early-stop threshold (val. loss)	0.025
Dropout (after Conv1)	0.4
Dropout (after GAP)	0.3
L2 regularization	0.002
CBAM reduction ratio r	8
Butterworth filter order	4
Random seed	42

3.4 Evaluation Metrics

Accuracy, weighted precision, weighted recall, weighted F1-score and Cohen's k are used to quantify the performance of the proposed model. Let TP_c, FP_c, FN_c and TN_c denote the true positives, false positives, false negatives and true negatives for class c, and let N be the total number of trials.

Accuracy measures the overall fraction of correctly classified trials:

$$\text{Accuracy} = \frac{\sum_c TP_c}{N} \tag{5}$$

Weighted Precision is the weighted average of per-class precision, weighted by class support n_c:

$$\text{Precision} = \sum_c \frac{n_c}{N} \cdot \frac{TP_c}{TP_c + FP_c} \tag{6}$$

Weighted Recall is the weighted average of per-class recall:

$$\text{Recall} = \sum_c \frac{n_c}{N} \cdot \frac{TP_c}{TP_c + FN_c} \tag{7}$$

Weighted F1-score is the harmonic mean of precision and recall, aggregated with class weights:

$$F_1 = \sum_c \frac{n_c}{N} \cdot \frac{2 \cdot TP_c}{2 \cdot TP_c + FP_c + FN_c} \tag{8}$$

Cohen's k accounts for chance agreement between the predicted and true labels:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{9}$$

where P_o is the observed accuracy and P_e is the expected accuracy under random labeling. A k of 0 indicates chance-level performance, while 1 indicates perfect classification.

Table 3. Per-subject classification metrics of MB-ATNet on BCIC-IV-2a.

Subject	Acc (%)	Precision	Recall	F1	k
S01	82.29	0.823	0.823	0.823	0.762
S02	58.76	0.582	0.588	0.579	0.449
S03	88.67	0.889	0.887	0.876	0.847
S04	82.99	0.834	0.830	0.830	0.773
S05	49.62	0.481	0.496	0.480	0.338
S06	68.90	0.694	0.689	0.698	0.584
S07	89.07	0.892	0.891	0.898	0.852
S08	88.43	0.887	0.884	0.886	0.844
S09	85.76	0.861	0.858	0.854	0.810
Mean	77.17	0.771	0.772	0.769	0.695

4 Results

4.1 Classification Performance

Table 3 reports the per-subject test-set classification accuracy and Cohen's k achieved by MB-ATNet on the BCIC-IV-2a dataset. Session T and Session E were recorded on separate days, introducing genuine cross-session variability due to electrode repositioning and fluctuations in subject mental state [2]; MB-ATNet is trained on Session T and evaluated on Session E without any session-specific fine-tuning. The model attains a mean accuracy of 77.17% with a standard deviation of 13.75% and a mean k of 0.696 ($\pm$ 0.181), with seven out of nine subjects exceeding 68% accuracy, confirming that the multiband spectral fusion strategy generalizes well across sessions. Peak performance is observed at 89.07% and 88.67% for S07 and S03 respectively. The relatively high standard deviation reflects well-documented inter-subject variability, where certain individuals exhibit weaker ERD/ERS modulation resulting in lower decoding accuracy [2, 9], as seen for S02 and S05 whose degradation is consistent with prior cross-session studies [3]. Overall, MB-ATNet learns discriminative spectral and temporal representations consistently across the majority of subjects under a fixed-parameter, subject-independent configuration.

4.2 Comparison with State-of-the-Art Methods

Table 4 presents a per-subject accuracy comparison of MB-ATNet against six MI-EEG classification methods on the BCIC-IV-2a dataset, covering publications from 2020 to 2023. The compared methods include PSCSP [6], Incep-EEGNet [4], TCNet [14], EEG-ITNet [3], TCNet-Fusion [2] and MO-CNN [7], spanning both classical signal-processing and deep learning approaches.

MB-ATNet achieves a mean accuracy of 77.17%, which is comparable to TCNet-Fusion (77.31%) and TCNet (75.77%), while outperforming Incep-EEGNet (73.31%), EEG-ITNet (73.15%), MO-CNN (71.72%) and PSCSP (74.39%). On

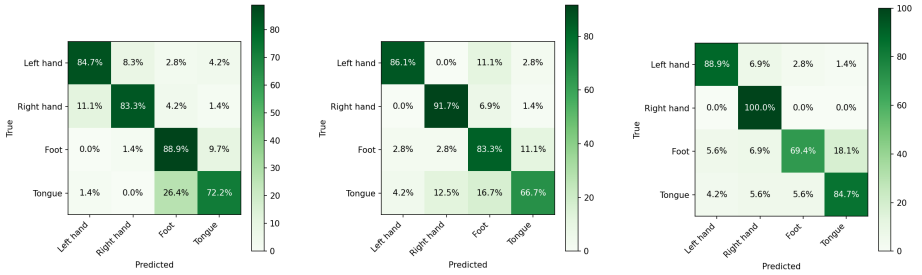


Fig. 2. Confusion matrices for subjects S01 (left), S04 (centre) and S09 (right). Rows represent true labels and columns represent predicted labels across four MI classes: left hand (LH), right hand (RH), foot (F) and tongue (T).

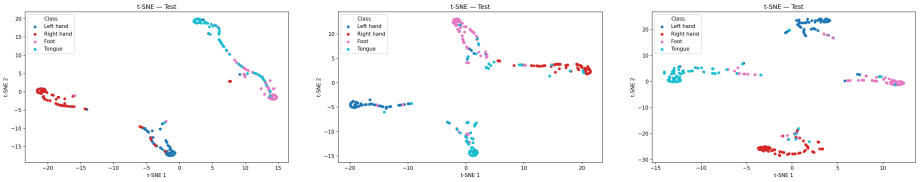


Fig. 3. t-SNE feature embeddings of MB-ATNet for subjects S01 (left), S04 (centre) and S09 (right), showing effective class discrimination.

subjects S04, S07, S08 and S09, MB-ATNet records accuracies of 82.99%, 89.07%, 88.43% and 85.76% respectively, surpassing all baseline methods on these subjects. This suggests that multiband spectral fusion coupled with CBAM attention effectively captures discriminative ERD/ERS patterns when inter-class separability is sufficiently present.

The lower performance on S02 (58.76%) and S05 (49.62%) is a trend consistently observed across all compared methods, indicating that these subjects pose a fundamental decoding challenge rather than reflecting a limitation specific to MB-ATNet. Notably, MB-ATNet achieves these results with only 26,234 parameters under a fixed, subject-independent training configuration, without any subject-specific hyperparameter tuning.

4.3 Discussion

MB-ATNet achieves a mean accuracy of 77.17% and a mean k of 0.696 across nine subjects on the BCIC-IV-2a dataset under a fixed, subject-independent configuration. The multiband spectral fusion strategy proves effective for subjects with well-pronounced ERD/ERS signatures, as evidenced by strong performance on S03, S07, S08 and S09, where MB-ATNet outperforms all six baseline methods.

The CBAM attention block aids inter-class discrimination by adaptively suppressing non-task-relevant frequency channels and temporal segments. The

Table 4. Comparison of classification accuracy (%) on BCIC-IV-2a.

Method	Year	S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg
PSCSP [6]	2020	80.00	65.36	87.14	67.50	55.54	50.18	91.79	84.11	87.86	74.39
Incep-EEGNet [4]	2020	84.72	54.86	86.81	70.83	44.42	54.86	92.71	87.85	82.64	73.31
TCNet [14]	2020	79.51	62.51	89.24	66.32	75.69	61.11	84.72	79.86	82.99	75.77
EEG-ITNet [3]	2022	77.78	63.19	83.68	76.39	72.22	61.11	73.26	75.69	75.01	73.15
TCNet-Fusion [2]	2021	82.64	62.85	90.62	67.36	76.04	64.24	87.15	82.29	82.64	77.31
MO-CNN [7]	2023	75.69	50.69	82.29	70.83	73.96	62.51	81.25	74.31	73.96	71.72
MB-ATNet (Ours)	2025	82.29	58.76	88.67	82.99	49.62	68.90	89.07	88.43	85.76	77.17

t-SNE visualizations for S01, S04 and S09 show well-separated class clusters in the learned feature space, confirming that the network extracts discriminative representations from the fused multiband inputs.

The weighted precision, recall and F1-score values in Table 3 closely track the accuracy trend across subjects, confirming consistent model behavior without class-specific bias. Subjects with strong alpha and beta ERD modulation, such as S03, S07, S08, yield F1-scores above 0.88, while S02 and S05 show lower scores consistent with their weaker rhythmic signatures. The most common misclassifications occur between foot and tongue MI classes, a pattern is consistently reported in the BCI literature [8, 9], while left-hand and right-hand classes are better separated due to their more spatially distinct cortical projections.

Subjects S02 and S05 yield the lowest accuracies of 58.76% and 49.62% respectively, consistent with BCI illiteracy, a condition in which individuals exhibit insufficient ERD/ERS modulation during motor imagery [2,3]. This trend is observed uniformly across all compared methods and addressing such subjects remains an open challenge in the field.

Table 4 places MB-ATNet among recent state-of-the-art methods. It outperforms PSCSP, Incep-EEGNet, EEG-ITNet and MO-CNN, while remaining competitive with TCNet [14] and TCNet-Fusion [2]. Notably, this performance is achieved with only 26,234 parameters, offering a favorable trade-off between accuracy and deployment efficiency.

Despite these results, several limitations should be acknowledged. The model is evaluated exclusively on BCIC-IV-2a without cross-subject or cross-dataset evaluation. Subject-specific hyperparameter tuning, known to yield accuracy gains, is not employed [2]. The gamma band is excluded due to susceptibility to muscular artifacts and subjects exhibiting BCI illiteracy remain challenging to decode under any fixed-parameter framework.

5 Conclusion

This paper presented MB-ATNet, a lightweight multiband attention network for four-class motor imagery EEG classification. The architecture combines raw and frequency-specific spectral streams (theta, alpha and beta) through early

concatenation, with a 1-D CBAM module providing joint channel and spatial recalibration. Evaluated on the BCIC-IV-2a dataset across nine subjects, MB-ATNet achieves a mean accuracy of 77.17% and a mean Cohen's k of 0.696, outperforming PSCSP [6], Incep EEGNet [4], EEG-ITNet [3] and MO-CNN [7], while remaining competitive with TCNet [14] and TCNet-Fusion [2]. With only 26,234 trainable parameters and no subject-specific tuning, MB-ATNet offers a favorable trade-off between classification accuracy and deployment efficiency, making it a suitable candidate for resource-constrained BCI applications.

Several directions present themselves for future investigation. Incorporating a gamma band stream with appropriate artifact suppression could capture the high-frequency motor signatures currently omitted. Subject-adaptive fine-tuning via meta-learning or transfer learning could address the performance gap for subjects with weak ERD/ERS modulation, such as S02 and S05. Evaluating MB-ATNet on additional benchmarks such as BCIC-IV-2b and OpenBMI [3] would further assess cross-dataset generalization and extending the framework to cross-subject and cross-session scenarios represents a practically important step towards clinically deployable BCI systems.

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